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Abstract

We address the EPIC-KITCHENS-100 Multi-Instance Retrieval (MIR) challenge through three progressively stronger approaches: (i) a JPoSE-based zero-fine-tuning baseline that exploits part-of-speech embeddings; (ii) a score-level ensemble of three pre-trained retrieval models (JPoSE, MI-MM and MMEN) with optional graph-diffusion re-ranking on visual and textual k -nearest-neighbour graphs; and (iii) end-to-end fine-tuning of an AVION ViT-L/14 dual encoder with the Symmetric Multi-Similarity (SMS) loss, which directly consumes the soft-label relevancy matrices, enhanced with horizontal-flip and longer-clip test-time augmentation. Across six iterations, average nDCG on the official Codabench leaderboard rises from 53.53 (JPoSE alone) to 55.82 (ensemble + re-ranking) and finally to 69.68 (AVION ViT-L + SMS, 16 epochs, with TTA), an absolute gain of +16.15 nDCG. We analyse where each gain originates, whether encoder capacity, soft-label supervision, test-time augmentation and additional epochs, and outline a model-soup-style ensemble of differently-trained ViT-L+SMS instances as future work, motivated by the compute and offline-evaluation constraints encountered.

1. Introduction

The growth of audiovisual content has driven the development of methods capable of analysing, indexing and automatically retrieving visual information. Within this landscape, multimodal video–text retrieval has consolidated as a relevant area at the intersection of computer vision and natural language processing. An especially challenging scenario is that of *egocentric video*, captured from a first-person perspective using head- or body-mounted cameras. Such recordings register natural interactions between hands, objects and the surrounding environment, introducing strong viewpoint, illumination and motion variability. Models must therefore interpret actions and contextual relationships at fine granularity in order to establish precise correspondences between visual content and textual descriptions.

The reference resource for this problem is the EPIC-KITCHENS-100 dataset [2], which contains over 100 h of egocentric kitchen video and approximately 90,000 narrated action segments. The associated Multi-Instance Retrieval (MIR) challenge formalises the task: given a textual query, the system must rank 9,668 trimmed video segments from the test split (and vice versa) by their semantic relevance, where ground-truth relevance is provided as a continuous soft-label matrix rather than as one-to-one binary correspondences [2, 16]. This formulation acknowledges that several segments may match a query at different degrees of similarity and so, is evaluated with order-sensitive metrics like *normalized discounted cumulative gain* (nDCG) [7] and *mean average precision* (mAP) rather than with the conventional Recall@K used in third-person retrieval. Despite recent progress in multimodal foundation models, semantic alignment between egocentric video and natural language remains difficult due to dynamic perspective changes, occlusions and multi-object interactions hinder the extraction of robust visual representations, and the fact that most existing losses do not exploit the graded relevance signal provided by the challenge.

2. Methodology

We approached EK-100 MIR as an iterative search through the design space, starting from a feature-based zero-fine-tuning baseline and ending with end-to-end fine-tuning of a strong egocentric backbone with soft-label supervision. The three approaches share a common formalization, stated once before each iteration is described.

2.1. Task formalization and common notation

Let $\mathcal{V} = \{v_i\}_{i=1}^{N_v}$ denote the $N_v = 9,668$ test video segments and $\mathcal{T} = \{t_j\}_{j=1}^{N_t}$ the $N_t = 3,842$ unique text queries on the test split. A retrieval system produces a similarity matrix $S \in \mathbb{R}^{N_v \times N_t}$ such that S_{ij} is large when v_i is semantically relevant to t_j . Ground-truth relevance is encoded in the soft-label matrix $R \in [0, 1]^{N_v \times N_t}$ [16] and the official challenge metric is the average normalized discounted

Text to video retrieval



Video to text retrieval

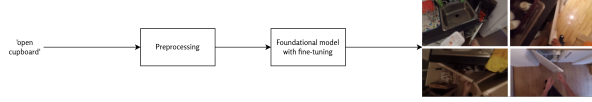


Figure 1. Overview of the multi-instance retrieval pipeline shared by the three approaches: visual and textual encoders project segments and queries into a shared space, and the resulting similarity matrix S is ranked and evaluated against the soft-label matrix R . Image components adapted from [2].

cumulative gain

$$\text{nDCG}_{\text{AVG}} = \frac{1}{2} (\text{nDCG}_{\text{V} \rightarrow \text{T}}(S, R) + \text{nDCG}_{\text{T} \rightarrow \text{V}}(S^{\top}, R^{\top})) \quad (1)$$

which captures both retrieval directions on equal footing. The three approaches differ only in how S is constructed while the evaluation protocol is identical across them. Figure 1 summarizes the resulting pipeline.

2.2. Approach 1: JPoSE Part-of-Speech baseline

Our entry point is a feature-based, zero-fine-tuning baseline using JPoSE [15]. It projects segments and queries into verb, noun and full-action sub-spaces, learned jointly with triplet supervision. Let $\phi_V^v, \phi_N^v, \phi_A^v$ and $\phi_V^t, \phi_N^t, \phi_A^t$ denote the visual and textual sub-encoders. We compute joint embeddings as the concatenation $\mathbf{f}_v(v_i) = [\phi_V^v(v_i); \phi_N^v(v_i); \phi_A^v(v_i)]$ and analogously $\mathbf{f}_t(t_j)$, and obtain $S_{ij}^{\text{JPoSE}} = \cos(\mathbf{f}_v(v_i), \mathbf{f}_t(t_j))$. We use the released JPoSE.BEST checkpoint without modification on pre-extracted features. By design this approach cannot exploit R and inherits the limitations of triplet supervision with binary positives and negatives, but it provides a reproducible reference point against which all subsequent iterations are compared.

2.3. Approach 2: score-level ensemble with graph-diffusion re-ranking

The second approach asks how far a purely inference-time strategy can carry the JPoSE baseline by combining several pre-trained retrieval models and refining the result with neighborhood-aware re-ranking. Three complementary models are queried independently: S^{JPoSE} as in Section 2.2; $S^{\text{MI-MM}}$, the multi-instance MaxMargin model of Miech et al. [10] with S3D-HowTo100M features at the released best epoch (202); and S^{MMEN} , an MLP-based Multi-Modal Embedding Network released as a baseline in [15].

Step 1: row-wise normalization and equal-weight fusion. The three matrices live on different scales (cosine similarities for JPoSE/MMEN, max-margin distances for MI-MM), so a direct sum is biased towards the highest-magnitude model. We min-max normalize each matrix *row-wise* so that, for every video query, scores against the 3,842 candidates are mapped to $[0, 1]$:

$$\tilde{S}_{ij}^{(m)} = \frac{S_{ij}^{(m)} - \min_k S_{ik}^{(m)}}{\max_k S_{ik}^{(m)} - \min_k S_{ik}^{(m)}} \quad (2)$$

$$S^{\text{ens}} = \frac{1}{3} (\tilde{S}^{\text{JPoSE}} + \tilde{S}^{\text{MI-MM}} + \tilde{S}^{\text{MMEN}}) \quad (3)$$

Equal weights are the natural default in the absence of a relevance-aligned validation signal.

Step 2: graph-diffusion re-ranking on visual and textual k-NN graphs. Inspired by diffusion-based re-ranking in image retrieval [6] and k-reciprocal encoding [19], we propagate ensemble scores through the local geometry of the retrieval space in both modalities. After L2-normalising each row of S^{ens} , a sparse visual affinity matrix $A_{vv} \in \mathbb{R}^{N_v \times N_v}$ stores cosine similarity between the score profiles of each pair of segments if either is among the top- k visual neighbors of the other. The same construction on $(S^{\text{ens}})^{\top}$ yields a textual affinity A_{tt} . We then expand the score matrix in both directions and blend with the original ensemble:

$$S^{\text{exp}} = \frac{1}{2} (A_{vv} S^{\text{ens}} + (A_{tt} (S^{\text{ens}})^{\top})^{\top}), \quad (4)$$

$$S^{\text{rerank}} = (1 - \alpha) S^{\text{ens}} + \alpha S^{\text{exp}} \quad (5)$$

with default $k = 20, \alpha = 0.3$. Both the ensemble matrix (3) and the re-ranked matrix (5) are submitted to Codabench as separate iterations, isolating the contribution of re-ranking.

2.4. Approach 3: AVION ViT-L fine-tuned with the SMS loss

The third approach replaces inference-time fusion with end-to-end fine-tuning of a strong CLIP-style egocentric backbone, supervised with the soft-label relevance matrix. The visual encoder is the AVION ViT-L/14 of Zhao and Krähenbühl [17], pre-trained on Ego4D via LaViLa [18] and the textual encoder is the CLIP text transformer paired with that AVION checkpoint. Both encoders are fine-tuned jointly with the Symmetric Multi-Similarity loss of Wang, Wang and Chau [13].

Symmetric Multi-Similarity loss. Within a training batch of B video-text pairs, let $\mathbf{x}_i, \mathbf{y}_i \in \mathbb{R}^d$ denote the L2-normalized visual and textual embeddings of pair i , $r_{ij} \in [0, 1]$ the soft relevance between v_i and t_j extracted from R [16], and $s_{ij} = \mathbf{x}_i^{\top} \mathbf{y}_j$. The SMS loss extends the

multi-similarity formulation of [11] so that, instead of binary pair weights, the contribution of each pair is modulated by r_{ij} . With margin θ , relevance threshold τ and temperatures β_p, β_n ,

$$\mathcal{L}_i^{V \rightarrow T} = \frac{1}{\beta_p} \log \left(1 + \sum_{j:r_{ij} \geq \tau} r_{ij} e^{-\beta_p(s_{ij}-\theta)} \right) + \frac{1}{\beta_n} \log \left(1 + \sum_{j:r_{ij} < \tau} (1 - r_{ij}) e^{\beta_n(s_{ij}-\theta)} \right) \quad (6)$$

and the full loss is the symmetric counterpart $\mathcal{L} = \frac{1}{2}(\mathcal{L}^{V \rightarrow T} + \mathcal{L}^{T \rightarrow V})$ averaged over the batch. Compared with InfoNCE-like contrastive losses, SMS pulls together pairs in proportion to their relevance and pushes apart pairs in proportion to $1 - r_{ij}$, providing a supervision signal directly aligned with the nDCG metric.

Fine-tuning configuration. Fine-tuning is launched from the AVION pretrain checkpoint via `amplus_finetune.py` from a patched fork of the public SMS-Loss codebase [12] on a single A100. Key flags are: `--model CLIP_ViTL14`, `--batch-size 48`, `--lr 2e-5`, `--loss-margin 0.6`, `--loss-thres 0.1`, with `--use-fast-conv1`, `--grad-checkpointing` (essential for ViT-L on a single GPU) and `--use-flash-attn` when available. Here, FlashAttention [3] roughly halves attention memory at no accuracy cost. We use AdamW for optimization [9] and one numbered checkpoint is saved per epoch. The relevance threshold $\tau = 0.1$ excludes pairs whose soft relevance is too small to provide a useful gradient and the margin $\theta = 0.6$ controls the separation enforced between positive and negative pairs. Both values are inherited from the public reference configuration of [13]. A trained model can be extended at any point by passing both `--pretrain-model` and `--resume` pointing to the same checkpoint and increasing `--epochs`, we use this hook to extend the base 10-epoch run to 16 epochs.

Divergences from the published recipe. Two configuration choices depart from the SMS-Loss recipe of Wang et al. [13] and shape the upper bound of what our pipeline can achieve. First, our effective batch size of 48 on a single A100 is approximately five times smaller than the 240 samples (60 per GPU $\times$ 4 GPUs) reported for ViT-L by the original authors. Smaller batches reduce the diversity of negatives visible to the SMS denominator in Eq. (6) and weaken the metric-learning signal. Second, our base run is restricted to 10 epochs (extended to 16 in iteration 6) against the 100 epochs of the published recipe. Both deviations are quantified in Section 5.2 and motivate the future-work direction in Section 5.3.

Inference and test-time augmentation. Inference is performed by `test_mir.py`, which encodes all test video segments and text queries and writes a similarity matrix in the same format as Approaches 1 and 2. We compare two settings: a *standard* pass with default 8-frame clips, and a *TTA* pass with `--flip --clip-length 32` that averages the embeddings of the original and horizontally-flipped clips and samples 32 frames per segment. Flip-averaging is the canonical video adaptation of [8] and exploits the near-invariance of kitchen activities to horizontal reflection; the longer clip length is motivated by the long tail of segment durations in Section ??.

3. Results

3.1. Evaluation protocol

All iterations are evaluated under a single protocol on the official EPIC-KITCHENS-100 MIR benchmark on Codabench¹. Each submission is a Codabench-compatible ZIP containing the $9,668 \times 3,842$ similarity matrix serialized as a protocol-2 pickle with a numpy namespace shim. It is scored by the grader against the held-out test relevance matrix [16] and the grader returns the average nDCG (Eq. 1) used to rank submissions. Because the test relevance matrix is not publicly released, we cannot compute mAP or per-direction nDCG offline. Therefore, we report the official Codabench leaderboard nDCG AVG as the single primary metric throughout the paper. As a sanity check on the implementation of Approach 1, the JPoSE codebase exposes an internal train-set validation routine on which we obtained an offline nDCG of 0.690 and mAP of 0.734, matching the figures of Wray et al. [15] before any submission was made. These offline numbers are not directly comparable to the public Codabench score for the same submission (53.53 nDCG) and are therefore not used for cross-iteration comparison.

3.2. Comparative results across iterations

Table 1 reports the official Codabench nDCG AVG for each of the six iterations described in Sections 2.2–2.4, together with the marginal contribution of every design change introduced over the previous iteration. The trajectory closes more than half of the gap between feature-based baselines and the current public state of the art (CR-CLIP at 82.10 nDCG [5]) on a single GPU and using only sixteen epochs of fine-tuning.

The breakdown delivers four observations. First, ensemble fusion of three frozen pretrained models contributes +1.78 nDCG, confirming that JPoSE, MI-MM and MMEN are complementary enough for score averaging to recover variance that none of them captures individually, but also

¹<https://www.codabench.org/competitions/12008/>

Table 1. Official Codabench nDCG AVG (%) across the six iterations of Section 2. Each row attributes a delta to the single design change introduced over the previous iteration. Best per approach in **bold**.

It.	Approach	Single change	nDCG AVG	Δ
1	A1 – JPoSE	Feature-based baseline (no fine-tune)	53.53	–
2	A2 – Ensemble	+ 3-model ensemble (no re-ranking)	55.31	+1.78
3	A2 – Ensemble	+ Graph-diffusion re-ranking ($k=20$, $\alpha=0.3$)	55.82	+0.51
4	A3 – AVION+SMS	Replace pipeline by AVION ViT-L + SMS, 10 ep	68.80	+12.98
5	A3 – AVION+SMS	+ TTA: <code>--flip --clip-length 32</code>	69.48	+0.68
6	A3 – AVION+SMS	+ Six additional fine-tuning epochs (10 $\rightarrow$ 16)	69.68	+0.20
<i>Public leaderboard SOTA, CR-CLIP [5]</i>			82.10	–
Total improvement, It. 1 $\rightarrow$ It. 6			–	+16.15

that the ceiling of frozen ensembles is modest in this setting. Second, graph-diffusion re-ranking adds +0.51 nDCG on top of the ensemble at no training cost, consistent with the magnitude of the gains historically reported in image retrieval [6]. Third, the single largest jump comes from replacing the inference-time pipeline with a single end-to-end fine-tuned AVION ViT-L supervised by SMS: +12.98 nDCG, dominating every other decision in the iteration plan by an order of magnitude and motivating allocating compute to encoder fine-tuning before exploring fusion strategies. Fourth, post-fine-tuning refinements behave as expected: TTA adds +0.68 nDCG with no training cost and extending fine-tuning from 10 to 16 epochs adds +0.20 nDCG. This suggests the model is approaching the plateau of single-checkpoint training which we revisit in Section 5.3 as motivation for a checkpoint-ensemble strategy.

4. Discussion

The per-iteration breakdown in Table 1 concentrates 80% of the total improvement (12.98 of 16.15 nDCG points) in the single transition from Approach 2 to Approach 3. That is from a frozen-feature ensemble to an end-to-end fine-tuned ViT-L driven by soft-label supervision. The asymmetry along the iteration axis is itself the central observation, and it can be unpacked along three connected lines: the low ceiling of inference-time refinements over frozen backbones, the dominant effect of the encoder-and-loss swap, and the diminishing returns of post-fine-tuning refinements that motivate the future-work direction.

Approach 2 fuses three pre-trained models that were never optimized for the soft-label setting: JPoSE and MMEN are triplet-supervised, MI-MM is binary multi-instance MaxMargin, and none sees the relevance matrix R during training. Equal-weight averaging of their normalized scores recovers +1.78 nDCG on top of JPoSE, in line with the canonical observation that retrieval ensembles reduce variance without correcting bias [14]. Because the three

models share the same failure mode (they ignore graded relevance), averaging them produces a more stable estimate of the same flawed objective rather than one that is closer to nDCG. Equal weights were used in the absence of a relevance-aligned validation signal and weighted averaging tuned on a held-out split could plausibly add a fraction of a point but is unlikely to overcome the bias ceiling. Graph-diffusion re-ranking adds another +0.51 nDCG at no training cost, consistent with magnitudes reported for diffusion in image retrieval [6]. A small $k = 20$ keeps the affinity graph sparse and avoids over-smoothing, moderate $\alpha = 0.3$ blends rather than replaces the original signal, and the symmetric construction in Eq. (??)—propagating through both visual and textual graphs—prevents either modality from dominating the result.

The single largest jump, +12.98 nDCG between iterations 3 and 4, is driven by three coupled changes that we deliberately bundled into a single iteration boundary by replacing pre-extracted features with an end-to-end ViT-L/14 encoder pre-trained on egocentric video [17, 18], replacing binary triplet/MaxMargin objectives by the SMS loss [13] that consumes the soft-label matrix R directly, and jointly fine-tuning both encoders rather than freezing them. Disentangling the relative contribution of each factor would require ablations beyond our compute budget. What is clear is that the gain dominates everything else by an order of magnitude. We attribute most of the effect to the combination of the strong egocentric backbone and the relevance-aware loss. The SMS objective in Eq. (6) weights both positive and negative pairs by their soft relevance, which aligns the gradient signal with the official nDCG metric (Eq. (1)) and is the only mechanism in our pipeline that directly optimises it. The size of the gain is consistent with the observation in [16] that graded-relevance objectives can reshape retrieval orderings well beyond what binary contrastive losses achieve.

Once the encoder has been fine-tuned end-to-end with SMS, however, further gains diminish quickly. TTA

(`--flip --clip-length 32`) yields +0.68 nDCG with no training cost. Flip-averaging adapts the canonical recipe of [8] to video and exploits the near-symmetry of kitchen scenes under horizontal reflection, while the longer clip length addresses the long tail of segment durations identified in Section ?? . Extending fine-tuning from 10 to 16 epochs adds a further +0.20 nDCG, suggesting the model is approaching the plateau of single-checkpoint training. This corresponds precisely to the regime in which model-averaging strategies tend to be most cost-effective and which motivates the future-work direction in Section 5.3. The practical implication of Table 1 is that, under a tight compute budget, encoder fine-tuning with relevance-aware supervision is the highest-yield investment by a wide margin: re-ranking and TTA are essentially free and should be applied by default, ensemble fusion of frozen baselines is also cheap but quickly hits a ceiling, and longer training is subject to rapidly diminishing returns. By contrast, the encoder swap returned +12.98 nDCG for the cost of a single A100 day. This ordering is consistent with what both the SMS-Loss authors [13] and the CR-CLIP authors [5] report on the same benchmark: the largest contribution to their final scores comes from the encoder choice and the relevance-aware loss, with subsequent refinements closing a smaller residual gap.

5. Conclusions, Limitations and Future Work

5.1. Conclusions

We presented a controlled, iterative study of the EPIC-KITCHENS-100 Multi-Instance Retrieval challenge in which three progressively stronger approaches were evaluated under a single Codabench protocol. Starting from a JPoSE feature-based baseline at 53.53 nDCG, a score-level ensemble of three frozen retrieval models combined with graph-diffusion re-ranking lifts performance to 55.82 nDCG, while an AVION ViT-L/14 dual encoder fine-tuned with the SMS loss and complemented by horizontal-flip plus 32-frame test-time augmentation reaches 69.68 nDCG over sixteen epochs of single-GPU training. The final iteration recovers more than half of the gap between feature-based baselines and the current public state of the art (CR-CLIP, 82.10 nDCG [5]) using a single A100 instance and the publicly released SMS-Loss training code. The per-iteration breakdown isolates where each gain originates. Encoder capacity coupled with relevance-aware supervision dominates by an order of magnitude, while inference-time strategies contribute small but reliable deltas at zero training cost. Additionally, this provides a transparent compute-allocation guideline for future EK-100 MIR work operating under similar resource constraints.

5.2. Limitations

Two limitations frame our results. First, all Approach 3 experiments were performed on a single NVIDIA A100 (40 GB) with a batch size of 48, which made multi-GPU runs, larger effective batch sizes and multi-seed re-training infeasible within the project timeline. Several knobs that are routine in the EK-100 MIR literature were therefore left unexplored. These include the contribution of larger batches (which directly affects the SMS gradient through the implicit hard-negative mining in Eq. (6)), systematic sweeps over θ , τ , β_p, β_n , longer fine-tuning schedules above 16 epochs, and variance estimation through repeated runs with different random seeds. Second, the EK-100 challenge releases the soft-label relevance matrix only for the training split. As mentioned previously, the test relevance matrix required to compute nDCG and mAP offline is held out by the organizers and is accessible exclusively through the Codabench grader. Model selection during fine-tuning therefore had to rely on training-loss trajectories rather than on a relevance-aligned validation signal, and every reported number in Table 1 required a Codabench submission, which capped the practical iteration budget. As a sanity check on the JPoSE baseline implementation we did obtain offline numbers on the train-set validation routine (0.690 nDCG, 0.734 mAP), but these are not directly comparable to the Codabench score for the same submission (53.53 nDCG) and were not used for cross-iteration comparison.

5.3. Future work

The most promising extension consistent both with the empirical evidence in Table 1 and with the SMS-Loss recipe of Wang et al. [13] is the *score-level ensemble of multiple SMS-supervised checkpoints* described in Section 3.3 of that paper but that we did not have the compute budget to reproduce. Concretely, the published recipe combines six independent models: three Adaptive MI-MM variants with different learning-rate schedules and three SMS variants with different relaxation thresholds ($\tau \in \{0.05, 0.10, 0.12\}$). They are all trained for 100 epochs from the same AVION pretrain [17] and combines them by directly summing their test similarity matrices, achieving an additional +1.1 mAP and +0.6 nDCG over the best single model. Replicating this ensemble on top of our AVION+SMS pipeline is a natural and well-defined extension due to the fact that row-wise normalization and equal-weight averaging used in Approach 2 (Eq. (3)) provide a drop-in replacement for the direct sum, and the graph-diffusion re-ranking of Eq. (5) can be stacked on top at no training cost. The rationale is supported by three observations from our trajectory and the related literature. First, longer training of a single ViT-L instance yields rapidly diminishing returns past 10–16 epochs, so additional compute is better invested in a few short complementary models than in extending a single one indefinitely. Second, model-soup-style strategies—in which several models fine-tuned with

different hyperparameters but the same initialization are combined—consistently match or exceed the best individual model in image classification [14] and have already been shown to work in this exact retrieval setting by [13]. Third, and our Approach 2 already validates the inference-time fusion machinery on weak frozen baselines, so deploying it on stronger fine-tuned ViT-L+SMS instances is essentially mechanical. The principal obstacle is computational: training six ViT-L+SMS instances for 100 epochs each requires roughly $40\times$ the GPU-hours of our iteration 6 and is not feasible on a single GPU within a typical project timeline. We therefore frame this direction as the natural continuation of the present work, conditional on multi-GPU infrastructure, in line with the 4×4 RTX 6000 Ada configuration used by Wang et al. for ViT-B and the 48 GB GPUs used for ViT-L. Beyond the ensemble direction, two further extensions follow from the limitations above: integrating the context-refinement mechanism of CR-CLIP [5] on top of our AVION+SMS pipeline (closing a documented portion of the gap to the public leaderboard), and replacing the SMS loss in Eq. (6) with a directly differentiable nDCG surrogate adapted to graded relevance (Smooth-AP [1] or a SoDeep-style ranker [4]).

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